

Threshold Strategy for Time-inconsistent Optimal Stopping Problems

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Optimal Stopping Problems

Consider the **price of an asset** evolves according to a stochastic process $X = (X_t)_{t \geq 0}$. The market allows holders to sell the asset at **any time**. The holder will receive a **payment** worth $g(X_\tau)$ if they sell at time τ .

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$$\sup_{\tau \in \mathcal{T}_t} \mathbb{E}_x \left[e^{-r(\tau-t)} g(X_\tau) \right], \quad (1)$$

where $\mathcal{T}_t$ is the set of all stopping times later than t .

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Due to the **exponential discounting function**, we can factor out e^{rt} , which is **independent** of τ , it does not change the optimal stopping rule. Hence, Problem (1) is **time-consistent**.

Discounting Function

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A more general and economically meaningful discounting function is the **weighted discounting function**:

Definition (Ebert, Wei, and Zhou 2020)

A weighted discounting function is a function $h : [0, \infty) \rightarrow (0, 1]$ such that $h(0) = 1$ and h is non-increasing, which explicit form is given by

$$h(t) = \int_0^{\infty} e^{-rt} dF(r),$$

where F is a distribution function on $[0, \infty)$.

Time Inconsistency I

Example

Consider a smoker with fixed lifetime $T > 0$ and a cost process

$$X_s = xe^{\frac{s-t}{2}}$$

in which $x > 0$, t is the current time and $s \in [t, T]$. The smoker faces two options:

- **Option 1:** Quit smoking at some time $s < T$ with a cost of X_s and die without any cost at time T .
- **Option 2:** Continue smoking until time T with a cost of X_T .

With the **hypobolic discounting function** $h(s) = \frac{1}{1+s}$, there is an optimal stopping problem

$$\inf_{s \in [t, T]} \mathbb{E}_x \left[\frac{X_s}{1+s} \right] = \inf_{s \in [t, T]} \frac{xe^{\frac{s-t}{2}}}{1+s}.$$

Time Inconsistency II

Solution

By basic calculus, we can conclude the optimal stopping time

$$s^*(t, x) = \begin{cases} t + 1 & \text{if } t < T - 1, \\ T & \text{if } t \geq T - 1. \end{cases}$$

Smoker will never quit smoking before death.

Definition (Huang and Nguyen-Huu 2018)

An optimal stopping problem is said to be time-consistent if the optimal stopping time

$$\tau_{t,x}^*(\omega) = \tau_{s, X_s(\omega)}^*(\omega)$$

for all $\omega \in \{\tau_{t,x}^* \geq s\}$ and $s \geq t$. Otherwise, the optimal stopping problem is time-inconsistent.

Ways to Handle Time Inconsistency

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- 1 Pre-committed method;
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- Game-theoretic approach finding subgame perfect Nash equilibrium;
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In this talk, we will focus on the game-theoretic approach.

Equilibrium Stopping

Denoting

$$J(t, x, \tau) = \mathbb{E}_x [h(\tau - t)g(X_\tau)];$$

$$\tau_u^t = \inf\{s \geq t : u_s(X_s) = 1\}.$$

Following the Notion provided by Ebert, Wei, and Zhou (2020)

Definition (Equilibrium Stopping Rule)

A stopping rule $\hat{u}$ is an equilibrium stopping rule if

$$\liminf_{\epsilon \rightarrow 0} \frac{J(t, x, \tau_{u^{\epsilon, a}}^t) - J(t, x, \tau_{\hat{u}}^t)}{\epsilon} \leq 0,$$

in which $u^{\epsilon, a}$ is the (ϵ, a) -deviation of $\hat{u}$ defined as follows:

$$u_s^{\epsilon, a}(x) = \begin{cases} \hat{u}_s & s \geq t + \epsilon, \\ a & s < t + \epsilon, \end{cases}$$

where $a \in \{0, 1\}$ and $\epsilon > 0$.

Spectrally Negative Lévy Processes

Theorem (Lévy-Itô Decomposition)

Given W is a standard Brownian motion, N is a Poisson random measure with intensity $ds\Pi(dy)$ and $\tilde{N}(ds, dy) = N(ds, dy) - ds\Pi(dy)$ is the compensated Poisson random measure, we can write X as

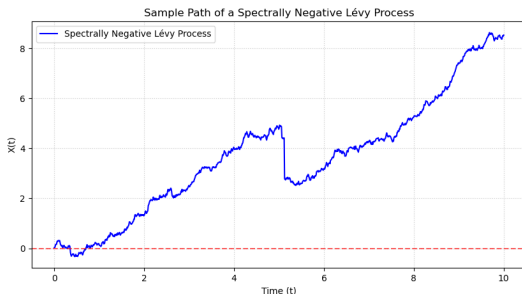
$$X_t = x + \mu t + \sigma W_t + \int_0^t \int_{(-\infty, -1)} y N(ds, dy) + \int_0^t \int_{(-1, 0)} y \tilde{N}(ds, dy),$$

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Main Results

Provided X is a spectrally negative Lévy process, it is natural to obtain candidate value function as

$$V(t, x) = \mathbb{E}_x \left[h(\tau_{\hat{u}}^t - t) g(X_{\tau_{\hat{u}}^t}) \right] = \mathbb{E}_x \left[h(\tau_{\hat{u}}^0) g(X_{\tau_{\hat{u}}^0}) \right] = V(0, x) = V(x).$$

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Assuming payoff function g is non-negative and non-decreasing with exponential growth, the candidate value function $V(x)$ will associate with a stopping time

$$\tau_{\hat{u}} = \inf\{t \geq 0 : X_t \geq w\}.$$

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Main Results

The candidate value function V is an equilibrium value function if and only if V satisfies the following conditions:

- $V(x) \geq g(x)$ for all $x < w$;
- $\mathcal{A}V(x) - \int_0^\infty rdF(r)g(x) \leq 0$ for all $x \geq w$;
- $V'(w-) = V'(w+) = g'(w)$.

Example I

Consider the payoff function with the following properties:

- non-decreasing, non-negative, and non-constant;
- its logarithm is concave;

Candidate value function as

$$V(x) = \begin{cases} g(x) & x \geq w, \\ \int_0^\infty e^{-\Phi(r)(w-x)} dF(r) g(w) & x < w, \end{cases}$$

where $\Phi(r)$ is the right inverse of the Laplace exponent $\psi(\theta) = r$ of X and w is smallest solution of the equation

$$\ln g(w) = \int_0^\infty \Phi(r) dF(r)$$

Defining

$$\hat{V}(x) = \begin{cases} g(x) & x \geq w, \\ e^{-\int_0^\infty \Phi(r) dF(r)(w-x)} g(w) & x < w. \end{cases}$$

Example II

Result

Applying the main results, if

$$\int_{-\infty}^0 \left[V(w + y) - \hat{V}(w + y) \right] \Pi(dy) \leq 0,$$

and Π is the Lévy measure of X with a completely monotone density π . Then V is an equilibrium value function.

Covering a lot of examples, including

- American Call Option;
- Real Option with Investment Cost;
- Power Utility Function.

Summary

- Study the time-inconsistent optimal stopping problem with weighted discounting function;
- Using the game-theoretic approach to find the equilibrium stopping rule and derive the necessary and sufficient conditions for the equilibrium value function;
- Solve the equilibrium value function and stopping rule for a class of payoff functions.

Thanks for your listening!
Any questions?

References I

-  Ebert, S., W. Wei, and X. Y. Zhou (2020). “Weighted discounting—On group diversity, time-inconsistency, and consequences for investment”. In: *Journal of Economic Theory* 189, p. 105089. DOI: 10.1016/j.jet.2020.105089. URL: <https://www.sciencedirect.com/science/article/pii/S0022053118305295>.
-  Huang, Y.-J. and A. Nguyen-Huu (2018). “Time-consistent stopping under decreasing impatience”. In: *Finance and Stochastics* 22.1, pp. 69–95.
-  Pedersen, J. L. and G. Peskir (June 2016). “Optimal mean-variance portfolio selection”. In: *Mathematics and Financial Economics* 11.2, pp. 137–160. ISSN: 1862-9660. DOI: 10.1007/s11579-016-0174-8. URL: <http://dx.doi.org/10.1007/s11579-016-0174-8>.
-  Strotz, R. H. (1955). “Myopia and Inconsistency in Dynamic Utility Maximization”. In: *The Review of Economic Studies* 23.3, p. 165. ISSN: 0034-6527. DOI: 10.2307/2295722. URL: <http://dx.doi.org/10.2307/2295722>.

References II



Weitzman, M. L. (Mar. 2001). “Gamma Discounting”. In: *American Economic Review* 91.1, pp. 260–271. DOI: 10.1257/aer.91.1.260.
URL:
<https://www.aeaweb.org/articles?id=10.1257/aer.91.1.260>.